

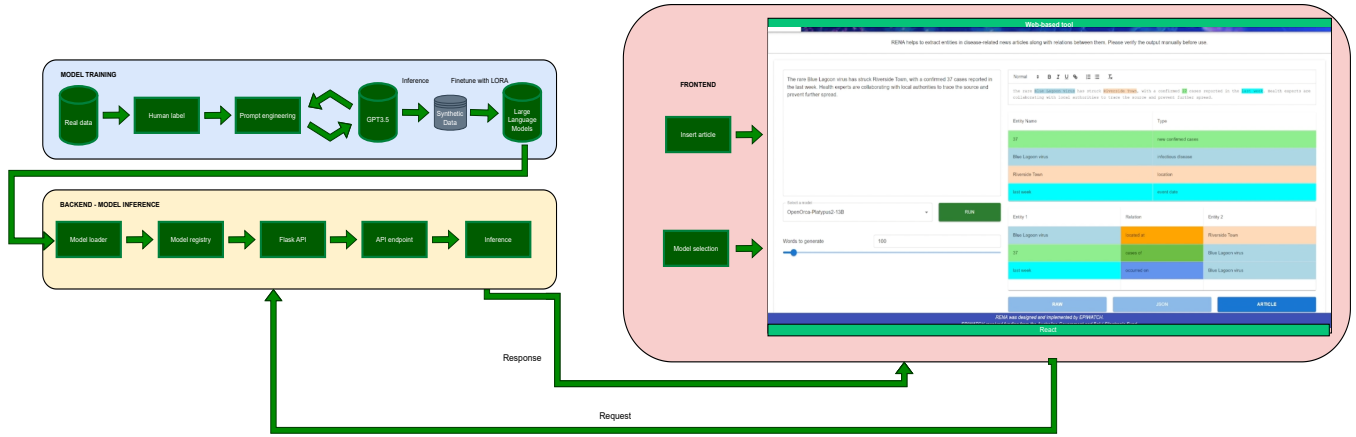


Figure 1: Architecture

| Model                  | Eval | Precision | Recall | F1   |
|------------------------|------|-----------|--------|------|
| Open-Orca-13b          | NER  | 0.93      | 0.66   | 0.77 |
| Open-Orca-13b          | RE   | 0.88      | 0.88   | 0.88 |
| Mythical-Destroyer-13b | NER  | 0.96      | 0.57   | 0.71 |
| Mythical-Destroyer-13b | RE   | 0.97      | 0.97   | 0.97 |

Table 1: Evaluation of models in RENA.

turbo-16k) (Details in Supplementary Document.). The generated output (articles, and labeled entities and relations) were used for fine-tuning two large language models including OpenOrca-Platypus2-13B<sup>3</sup> and Mythical-Destroyer-V2-L2-13B<sup>4</sup>, both based on Meta’s LLaMa2 13B model (?), using quantized low-rank adaptation (QLoRa) (?). We run the training process for three epochs. In the inference phase, RENA allows for the selection of the two different models specified above. Our backend is built with Python Flask, where the model is both loaded and executed for inference. The output is then sent to our frontend, developed in React, to display the response.

We evaluated the two models included in RENA on 10 randomly selected articles in English, selected from the EPI-WATCH database, an existing database with more than 1 million news articles relating to infectious disease outbreaks searched through online news, and prioritised using machine learning-based classification supplemented with manual curation by experts. We computed precision, recall and F1 score, comparing the predicted to labeled output on the ten articles for each model. We assessed the accuracy for named entity recognition (NER) and RE separately. RE was only evaluated for relevant entities that were recognised, so both metrics should be considered when assessing the models. Table 1 shows the precision, recall and F-score values for the models included in RENA.

<sup>3</sup><https://huggingface.co/Open-Orca/OpenOrca-Platypus2-13B>; Accessed on 3rd October, 2023.

<sup>4</sup><https://huggingface.co/Sao10K/Mythical-Destroyer-V2-L2-13B>; Accessed on 3rd October, 2023.

## User Interface

Figure 1 is a snapshot of the web interface hosting RENA. On the webpage includes:

- **Input:** The input contains the text field that the user can input their unstructured article into, from which the model will extract entities and relations.
- **Select Model:** The select model box contains a drop-down to select between the 2 models to be used for generation.
- **Submit:** Begins the generation process.
- **Max tokens:** Max tokens is a slider to determine how many tokens of output will be produced by the model.
- **Output:** The model will generate the output (relations) into this box.
- **Raw Button:** The raw generated output of the model will be displayed in the output box when clicked.
- **JSON Button:** The output will be converted into a JSON format and be displayed in the output box when clicked.
- **Article Button:** The article will be displayed in the output box and the entities will be highlighted in the text and color coded. A table showing all the entity and entity types as well as another table showing the relations will also be shown.

## Summary & Future Work

RENA is a tool that uses QLoRA to finetune LLMs and extract disease-related relations from news articles. It offers a way to automate the creation of structured data from an inputted unstructured news article and output relatively accurate results. RENA will enable users interested in researching diseases to parse large amounts of data to potentially aid in the surveillance and prevention of diseases. We plan to extend RENA to multilingual news articles. In addition, different relations can help delineate different disease events, allowing detection of independent outbreaks involving different infections or locations from news articles. Nostrum a sequi qui necessitatibus ex dolore, delectus suscipit quasi fugiat, voluptatum illum nobis beatae ea hic eligendi tempore

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